Egg Packing Machine



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Egg Packing Machine

What it is used for

An egg packing machine sorts, orients, and places eggs into trays, cartons, or other packaging according to the machine design. Correct setup, gentle handling, and routine checks help reduce cracked eggs, contamination, and packaging errors.

Main parts

- Egg infeed conveyor or loading table
- Egg orientation and transfer mechanism
- Grading or sorting section where fitted
- Tray or carton feed system
- Packing conveyor and collection area
- Control panel, guards, sensors, and drive system

Before you start

1. Confirm the egg size range, packaging type, batch requirements, and machine capacity.
2. Inspect conveyors, rollers, guides, cups, sensors, guards, and drive components for damage or residue.
3. Verify that food-contact surfaces and egg-contact parts are clean and suitable for use.
4. Check that trays and cartons are clean, dry, undamaged, and appropriate for the product.
5. Confirm that emergency stops and guards are in place and that the operating area is clear.
6. Follow hygiene procedures, wear clean protective clothing, and check that the machine is correctly assembled.

How to use

1. Switch off and isolate power before installing or adjusting removable guides, cups, or packaging components.
2. Fit the required tray or carton format and adjust guides according to the manufacturer's instructions.
3. Check that the infeed path, transfer mechanism, and collection area are clear of obstructions.
4. Switch on the machine and select the approved operating settings for the egg and packaging format.
5. Feed eggs gently and evenly into the designated infeed area without dropping or crowding them.
6. Start at a controlled speed and observe the transfer of eggs through orientation and packing stages.

7. Check that eggs are seated correctly in the trays or cartons and that packaging remains aligned.
8. Inspect early output for cracked eggs, empty positions, misplaced eggs, and damaged packaging.
9. Adjust approved settings or reduce feed rate if breakage, misalignment, or jams occur.
10. Monitor the machine during operation and remove damaged eggs using the approved hygienic procedure.
11. Stop the machine immediately if guards fail, an unsafe jam occurs, or abnormal noise or movement is detected.
12. Isolate power and wait for all moving parts to stop before clearing jams or reaching into the mechanism.
13. At the end of the batch, stop feeding eggs, complete the approved clearing sequence, and switch off the machine.
14. Remove packed products carefully, record batch and quality information, and clean egg-contact surfaces after use.

⚠ Safety

- Only trained personnel should operate or adjust the egg packing machine.
- Keep hands away from moving conveyors, rollers, transfer cups, and pinch points.
- Never bypass guards or reach into the machine while it is running.
- Switch off and isolate power before clearing jams, changing formats, or performing maintenance.
- Handle broken eggs promptly using the site's hygiene and waste procedures to prevent contamination and slipping hazards.
- Keep water and cleaning liquids away from electrical controls unless the equipment is designed for the approved cleaning method.
- Follow site biosecurity and food-safety procedures to prevent contamination between batches.

Care & storage

- Clean egg-contact surfaces, guides, conveyors, and collection areas according to the approved sanitation schedule.
- Inspect cups, rollers, belts, sensors, and guides for wear, damage, or misalignment.
- Check fasteners, guards, and emergency stops at the prescribed intervals.
- Keep electrical components protected during cleaning and follow the manufacturer's instructions.
- Arrange servicing if egg breakage rises, sensors malfunction, or the transfer mechanism becomes unreliable.

Common mistakes to avoid

- Feeding eggs too quickly or dropping them onto the conveyor.
- Using trays or cartons that do not match the selected machine format.
- Ignoring cracked eggs, misalignment, or repeated jams.
- Clearing an obstruction before the machine is isolated and stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.